ORCA — Marine EcOsystem Reasoning with Collaborative Agents

Master Presentation Script & Stage Delivery Guide (6-Minute SIH Pitch)

Problem Statement ID: SIH26176	**Team Name: CodeCatalysts**
Pitch Duration: 6 Minutes Strict (1 min / slide)	**Jury Focus: Technical Rigor, Grounding, Live Impact**

■ EXECUTIVE STAGE RULES FOR TEAM CODECATALYSTS

• **Time Management:** 60 seconds per slide. Do not dwell on generic problems — get to ORCA's 8 agents and live results by Minute 2.
• **The Core Defense Soundbite:** *'ORCA uses an LLM solely for intent understanding and conversational synthesis. All ocean measurements, risk scores, routes, and coordinates are computed deterministically with zero data hallucination.'*
• **Speaker Handoffs:** Speaker 1 (Captain / Lead) opens and closes. Speaker 2 (Tech Architect) covers the 8 Agents & Mathematical Engines. Speaker 3 (Domain & Operations Lead) handles Feasibility, Impact & Verification.

SLIDE 1: TITLE PAGE: Problem Statement SIH26176

■■ **Target Duration: 0:00 – 0:35 (35 Seconds)** ■■ **Allocated Presenter: Speaker 1 (Team Lead / Captain)**

■ ***Visual Cues & Stage Choreography:***
Slide 1 is visible on screen with SIH logo, Team CodeCatalysts, and the glowing ORCA AI lightbulb graphic.

Spoken Speech (Word-for-Word Narration):

"Respected jury members and esteemed evaluators, good morning. We are Team CodeCatalysts, and today we are thrilled to present our working solution for Problem Statement SIH26176: **ORCA — Marine EcOsystem Reasoning with Collaborative Agents**.

India possesses over 11,000 kilometers of coastline, where more than 28 million fishermen and coastal stakeholders venture into the sea daily. Yet, their operational safety and livelihoods remain hostage to fragmented bulletins, complex satellite maps, and static weather PDFs. Today, we bridge this gap with ORCA — India's first agentic AI marine intelligence copilot that transforms scattered oceanographic data into life-saving, real-time decisions."

■ **Key Memorized Punchline: Key Soundbite: 'We don't need more scattered marine dashboards; we need a collaborative agentic brain that reasons across them.'**

SLIDE 2: PROPOSED SOLUTION & DOMAIN RELEVANCE

■■ Target Duration: 0:35 – 1:40 (65 Seconds) **■■ Allocated Presenter: Speaker 1 (Team Lead)**

■ ***Visual Cues & Stage Choreography:***
[TRANSITION TO SLIDE 2] Point clearly to the 3 top pillars: Detailed Solution, Problem Resolution, and Innovation. Gesture to the bottom stats.

Spoken Speech (Word-for-Word Narration):

"Moving to Slide 2: What exactly is ORCA?

ORCA acts as an intelligent maritime copilot. A fisherman or vessel master simply asks in plain language — in English, Tamil, Malayalam, Hindi, or Marathi — *'Is it safe to fish offshore Kochi tomorrow, and where are the nearest tuna hotspots?'*

Instantly, ORCA decomposes this query across **8 Collaborative AI Agents**, pulling live observations from NOAA GHRSST satellite radiometry, ISRO Oceansat-3 chlorophyll, Open-Meteo marine models, and INCOIS deep-sea OMNI buoys.

Why does this matter so critically right now? Look at the numbers on screen: India's seafood exports have crossed ■62,400 crore, and 95% of our international trade moves across these waters. Yet, artisanal fishermen burn up to 30% of their earnings on wasted diesel searching for fish in barren or dangerous waters. ORCA provides pinpoint Potential Fishing Zones with precise compass bearings, predicts fuel savings, and delivers strict Sail/No-Sail safety clearances before the vessel unmoors."

■ Key Memorized Punchline: Key Soundbite: 'ORCA cuts voyage fuel burn by 20–30% while guaranteeing that no vessel unwittingly sails into a cyclone or naval restricted zone.'

SLIDE 3: TECHNICAL APPROACH & AGENT ARCHITECTURE

■■ **Target Duration: 1:40 – 3:00 (80 Seconds)** ■■ **Allocated Presenter: Speaker 2 (Technical Architect)**

■ ***Visual Cues & Stage Choreography:***
[TRANSITION TO SLIDE 3] Direct judges' attention to the Implementation Flow (Steps 1–7) on the left, then the 8 Specialized Agents in the center.

Spoken Speech (Word-for-Word Narration):

"Thank you. On Slide 3, we present the technical backbone of ORCA. Our architecture operates on a strict 7-step execution pipeline:

When a query arrives, it first hits our **Planning Agent**. Rather than relying on an LLM to hallucinate answers, the Planning Agent acts purely as an orchestrator — decomposing intent, timeframe, and resolving coordinates against our pre-indexed 35+ coastal port registry.

It then triggers only the specialized agents needed:
• **Marine Retrieval & Ocean Analytics Agents** compute thermal gradient fronts (ΔT/km) and chlorophyll upwelling.
• **Weather Intelligence Agent** derives wind vectors, Douglas Sea State (0–9), and Beaufort Force (0–12).
• **Alert Agent** parses live INCOIS High Wave alerts and IMD Port Danger Signals 1 through 11.
• **Risk Assessment Agent** runs a deterministic multi-parameter scoring algorithm (0–100) — zero LLM guesswork.
• **Geospatial Agent** calculates dual routes: Route Alpha along standard corridors, and Route Bravo deep-water bypass that rounds Cape Comorin with **0% land crossing**.
• Finally, the **Response Synthesis Agent** produces a fully cited report with complete Data Provenance ledgers.

Our stack uses Python 3.13, FastAPI, Supabase PostGIS with an embedded ray-casting spatial index running sub-5ms queries, and a custom 60 FPS HTML5 Canvas fluid simulator."

■ **Key Memorized Punchline: Key Soundbite: 'The LLM understands the sailor's question; the deterministic scientific engines compute the truth.'**

SLIDE 4: FEASIBILITY & VIABILITY (RISK MITIGATION)

■■ **Target Duration: 3:00 – 4:05 (65 Seconds)**	■■ **Allocated Presenter: Speaker 2 / Speaker 3 (Technical & Operational Lead)**

■ ***Visual Cues & Stage Choreography:***
[TRANSITION TO SLIDE 4] Emphasize the 8 Challenges vs Solutions table on the right side of the slide.

Spoken Speech (Word-for-Word Narration):

"On Slide 4, we address the hardest question in mission-critical marine software: *Is this truly feasible and viable at sea?*

Our MVP is not a mock-up — it is backed by a **100% passing automated test suite of 34 rigorous pytest cases** covering every SIH scenario.

Let us highlight how we overcome the core challenges:
1. Data Freshness: We enforce an automated freshness validator. If satellite or weather data is older than 12 hours, an explicit **+8 uncertainty risk penalty** is added to the safety score.
2. Zero-Hallucination: LLMs never output synthetic measurements. All numbers are pulled directly from live APIs and verified formulas.
3. Maritime Boundaries & IMBL: When a vessel approaches within 12 nautical miles of the Sri Lankan or Pakistani International Maritime Boundary Line, our PostGIS geofence fires an instant visual and audio alarm.
4. Connectivity Loss: Our PWA architecture caches static nautical charts and pre-seeded bathymetry soundings via Service Workers, allowing full offline operation when offshore cellular drops."

■ **Key Memorized Punchline: Key Soundbite: 'We don't sweep edge cases under the rug — our system explicitly calculates uncertainty margins for delayed telemetry.'**

SLIDE 5: IMPACT, BLUE ECONOMY BENEFITS & SCALABILITY

■■ Target Duration: 4:05 – 5:10 (65 Seconds)	■■ Allocated Presenter: Speaker 3 (Domain & Blue Economy Lead)

■ ***Visual Cues & Stage Choreography:***
[TRANSITION TO SLIDE 5] Walk across the 5 stakeholder impact cards, highlighting the tangible economic and ecological numbers.

Spoken Speech (Word-for-Word Narration):

"Slide 5 illustrates the multi-dimensional impact of ORCA across India's Blue Economy:

• **For Artisanal Fishermen:** We eliminate blind trawling. By guiding craft to active thermal convergence zones, we save **150 to 250 liters of diesel per voyage**, equating to ■15,000 in direct profit increase per trip while slashing rough-sea capsizing.
• **For Government & Coast Guard:** ORCA includes an integrated **GMDSS Mayday Emergency SOS System**. In case of engine failure or hull breach, one tap dispatches an authenticated distress telegram with coordinates to MRCC Kochi, Mumbai, or Chennai, calculates SAR intercept ETA, and generates an IMO voice script.
• **For Environmental Sanctuaries:** Our spatial engine automatically routes vessels outside sensitive marine sanctuaries like **Gahirmatha** (protecting Olive Ridley turtle nesting) and the **Gulf of Mannar Biosphere**.

Scalability: Tested across the West and East coasts, our Dockerized cloud-native microservices scale effortlessly from coastal pilot to all 72 maritime districts."

■ Key Memorized Punchline: Key Soundbite: 'ORCA aligns directly with the Prime Minister's Matsya Sampada Yojana (PMMSY) and Sagarmala — powering coastal security and prosperity.'

SLIDE 6: RESEARCH, DATASETS & OFFICIAL CITATIONS

■■ **Target Duration: 5:10 – 6:00 (50 Seconds)**	■■ **Allocated Presenter: Speaker 1 (Team Lead / Conclusion)**

■ ***Visual Cues & Stage Choreography:***
[TRANSITION TO SLIDE 6] Point to the authoritative bodies: INCOIS, ISRO, IMD, Coast Guard, NOAA.

Spoken Speech (Word-for-Word Narration):

"Finally, Slide 6 validates our research foundation and data provenance:

ORCA relies on peer-reviewed operational oceanography — Douglas Sea State scales, Beaufort wind forces, and ISRO/INCOIS thermal front edge detection models. Every single recommendation embeds verifiable data citations from **ISRO MOSDAC, NOAA CoastWatch GHRSST, ECMWF, IMD, and the National Data Buoy Programme**.

In summary: ORCA is reliable, fully functional, multi-agent grounded, and engineered for India's maritime future. We now invite the honorable judges to witness our live system demonstration. Thank you!"

■ Key Memorized Punchline: Final Winning Call: 'ORCA: Where Agentic AI meets Maritime Truth. We are ready for your questions and demo!'